

Scalability & Future plan

Date	12 July 2026
Team ID	XXXX
Project name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum marks	1 mark

Scalability & Future plan

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Predictions depend only on the trained dataset	Accuracy may reduce for unseen or changing agricultural conditions	High
2	No real-time weather data integration	Crop recommendations may not reflect current weather conditions	High
3	Works only on a local Flask server	Farmers cannot access the application remotely	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	Users	Supports local users	Deploy on cloud to support multiple users simultaneously
2	Data Storage	CSV dataset	Use MySQL or MongoDB for efficient data management
3	Performance	Local Flask application	Optimize model and deploy using cloud services (AWS/Azure/GCP)
4	Security	Basic input validation	Add user authentication, HTTPS, and secure database access

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve crop prediction accuracy using advanced ML algorithms	Next 2 Months	More accurate crop recommendations

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time weather and soil data APIs	Next 4 Months	Better recommendations based on live environmental conditions
Phase 3	Deploy the application on a cloud platform and add user accounts	Next 6 Months	Improved accessibility, scalability, and security
Phase 4	Develop a mobile application with multilingual support	Next 12 Months	Increased usability for farmers across different regions